

The empirical recurrence for OEIS A266057, recovered from its tail

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8 September 2026

Abstract

OEIS A266057 records “Empirical recurrence of order 94” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 219$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 100$. Its last coefficient is $c_{94} = -915404976 \neq 0$, so no further tail satisfies anything shorter; any order-94 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A266057 is “Number of $3 \times n$ integer arrays with each element equal to the number of horizontal and antidiagonal neighbors less than or equal to itself.”. It has offset 1 and begins

1, 35, 20, 302, 10310, 127492, 1590691, 20375630, ...

The entry states:

Empirical recurrence of order 94 (see link above)

As of the “Last modified” line on the live entry (Dec 20 2015 10:28:16, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 219$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 219$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 94: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 94.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 438$ exact terms from $n = 6$ on, it returns order 94 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{94} c_i a(n-i),$$

c_1	2	c_{25}	−20626697521939473	c_{49}	−77541328220355669664057	c_{73}	1004868747
c_2	20	c_{26}	−150092829091187155	c_{50}	−10568952420310596853915	c_{74}	−31916112
c_3	1082	c_{27}	−230984250695649871	c_{51}	180126550534366807839778	c_{75}	−20743422
c_4	6427	c_{28}	195046535084717135	c_{52}	−140074303283317896344836	c_{76}	61674374
c_5	41965	c_{29}	1248566260955961772	c_{53}	−21864971650930685730778	c_{77}	−6759791
c_6	−47556	c_{30}	863086874884784391	c_{54}	304998662211746626646680	c_{78}	−4885694
c_7	−1500878	c_{31}	−5474221216053186204	c_{55}	−120208257752618435412507	c_{79}	14745198
c_8	−9598498	c_{32}	−18729317926834343237	c_{56}	−40465966109161329706424	c_{80}	34777981
c_9	−1240052	c_{33}	−29218800396413674280	c_{57}	254047592700577651316613	c_{81}	−2278894
c_{10}	248174558	c_{34}	−22688243038417196959	c_{58}	−133342675419908744773204	c_{82}	−5089118
c_{11}	1078439197	c_{35}	17535601846152982678	c_{59}	−80125582985246425172226	c_{83}	24004467
c_{12}	−2126499377	c_{36}	198831986818837634579	c_{60}	142026970394763909115649	c_{84}	8584524
c_{13}	−32779965271	c_{37}	796733663640599168176	c_{61}	−109048263039140935258938	c_{85}	7047967
c_{14}	−85936942577	c_{38}	1960815043644076032147	c_{62}	−40909398698788750350312	c_{86}	−277662
c_{15}	283299707550	c_{39}	3099893676079821719500	c_{63}	75722139217167333849178	c_{87}	−78761
c_{16}	2730272374332	c_{40}	2336376182755139663302	c_{64}	−30422159904285337348821	c_{88}	−30325
c_{17}	7157674196156	c_{41}	−240278137677506036227	c_{65}	−14351035331809561122128	c_{89}	18438
c_{18}	−8787739440517	c_{42}	−9311639271333110279741	c_{66}	23803739483659909953008	c_{90}	48466
c_{19}	−130734418666760	c_{43}	−10952187314821225269165	c_{67}	990800922084636213390	c_{91}	−4741
c_{20}	−429132569685856	c_{44}	−6138460323356718130674	c_{68}	−9267067909335289798744	c_{92}	−1963
c_{21}	−305344617364412	c_{45}	−10133664205475116233826	c_{69}	565280651167538988627	c_{93}	−187
c_{22}	2873670585120779	c_{46}	−18801345027023739258419	c_{70}	3272074262359851702200	c_{94}	−91
c_{23}	12391369113068518	c_{47}	21861646208866026166678	c_{71}	−2169021114713757231435		
c_{24}	18953468317197405	c_{48}	51315382674101671864002	c_{72}	−292282810757091905110		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 100$, and it is the recurrence OEIS A266057 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{94} - c_1x^{94-1} - \dots - c_{94}$. Its constant term is $-c_{94} = 915404976$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 94 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 94, so $\deg q = 94$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 94, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 94, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -915404976 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-94 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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